

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	KE + gravitational PE = 0 & context explained briefly (also accept KE = gravitational PE and context) (1) Minimum context = conservation of energy or mention of galaxy or spherical universe or uniform universe or cosmological principle or explaining why $r = D$ or explaining what the mass of the sphere is (accept whole universe or part and mass inside) Reference to universe nor expanding now loses 1 st mark $v = H_0 D$ substituted for v (1) mass = density $\times$ volume of sphere (1) Algebra set out clearly (1)	4			4	3	
		(ii)	Correct substitution into formula (gives 8.66×10^{-27}) (1) Dividing by 1.66×10^{-27} (or 5.2 or 5.22 seen) OR $5u = 8.3 \times 10^{-27}$ (1)	1	1		2	1	
	(b)	(i)	Halfway between or correct substitutions into C of M equation leading to 1×10^{11} or simply 1×10^{11} [m]		1		1		
		(ii)	Substitution into period equation (1) 9.73×10^6 s (allow 1 mark for 13.8×10^6 OR 4.87×10^6 OR 3.44×10^6) (1)	1	1		2	2	